

Eco Planet Solar

Full-Stack E-Commerce Platform

Complete Feature Reference · May 2026

This document lists every feature shipped across the Django + Next.js production platform. Features are grouped by domain for quick reference.

1. Platform Overview

- Full-stack production e-commerce platform for solar energy products and services
- Backend: Django 5 · Django REST Framework · SimpleJWT · Celery · Redis · PostgreSQL
- Frontend: Next.js 14 (App Router) · TypeScript · Tailwind CSS · Zustand · React Query
- Payments: Stripe (card, UPI) · PayPal v2 · Cash on Delivery
- Storage: Local FileSystem (dev) · AWS S3 via django-storages (production)
- Email: SMTP (SES / Postmark / any provider) with Celery-backed async dispatch
- 131 automated backend tests passing · full TypeScript strict build

2. Authentication & User Management

Registration & Login

- Email + username registration with duplicate detection
- JWT (Access + Refresh) authentication via `/api/auth/login/` and `/api/auth/token/refresh/`
- Secure logout invalidating the refresh token server-side
- Rate-limited login endpoint (`AuthRateThrottle`) to prevent brute-force attacks

Password Management

- Change-password endpoint for authenticated users (old + new required)
- Forgot-password flow: secure token emailed, 15-minute expiry
- Reset-password endpoint validates token + sets new password

Profile

- PATCH `/api/auth/profile/` — update first/last name, phone
- Email is immutable (change-email flow not exposed to prevent takeover)
- `is_staff` and `is_installer` flags exposed as read-only (admin-managed only)

Addresses

- Full CRUD for shipping & billing addresses
- Supports shipping / billing address types
- Default address flag per user
- Validated country / state / postal-code fields

3. Product Catalogue

Products

- Full product model: name, slug, SKU, description, technical description
- Category hierarchy (parent/child categories)
- Pricing: base price + optional discount percentage != computed discounted_price
- Capacity, brand, warranty years, lifespan years fields
- Installation: is_available flag + installation_fee amount
- Stock tracking (integer) with low-stock threshold alerts
- Delivery days estimate
- Tags field for flexible filtering
- Active / inactive toggle for catalogue management
- Featured flag surfaced on storefront homepage

Product Images

- Multiple images per product
- Primary image flag
- Alt-text and sort-order for accessibility and ordering
- S3 / local storage depending on USE_S3 environment variable

Filtering & Search

- Full-text search across name, brand, category
- Filter by category slug, is_featured, price range
- Ordering by price (asc/desc) and created date
- Paginated results (cursor / page-based)

CSV Import / Export (Admin)

- GET /api/products/export-csv/ — streaming CSV export of entire catalogue
- POST /api/products/import-csv/ — multipart CSV upload with upsert by SKU
- Returns per-row created / updated / skipped / errors counts
- Handles utf-8-sig encoding (Excel compatibility)

Related Products

- GET /api/products/<slug>/related/ — products in the same category
- Featured products endpoint for homepage hero carousel

4. Shopping Cart

- Server-side cart stored per authenticated user
- Add / update quantity / remove line items
- Installation add-on per line item (checkbox)
- Cart totals: subtotal, installation total, item count
- Stock validation on add (cannot exceed available stock)
- Cart cleared automatically on successful order placement
- Cart badge in header updates in real-time via React Query

5. Wishlist

- Server-side wishlist per authenticated user

- Add / remove products
- Toggle wishlist button on every product card
- Wishlist count badge in header
- Full wishlist page with "Add to cart" shortcut
- Persists across devices/sessions

6. Checkout & Orders

Checkout Quote (live preview)

- POST `/api/orders/checkout/quote/` — debounced live breakdown as user fills form
- Returns: subtotal, installation, discount, tax, shipping cost, ETA range, grand total
- Coupon validation included in quote

Order Placement

- Select shipping address from saved addresses
- Select payment method: Stripe / PayPal / COD
- Apply coupon code (validated server-side)
- Tax computed server-side at configurable rate (default 18% GST)
- Shipping cost computed via zone-based shipping rules
- Grand total = subtotal + installation + discount + tax + shipping
- Optional order note field

Order Management (Customer)

- List own orders with status and payment filters
- Order detail page with full item breakdown and totals
- Cancel own pending order with reason
- Track shipment with courier, tracking number, ETA, and external link
- Stripe payment block on unpaid orders (inline card/UPI)
- PayPal capture block on unpaid PayPal orders
- Request a return (after delivery)

Order Statuses

- pending !' confirmed !' processing !' shipped !' delivered / cancelled
- Status transitions enforced server-side (no arbitrary jumps)

Payment Statuses

- unpaid · paid · refunded

7. Payment Integrations

Stripe

- PaymentIntent with automatic_payment_methods (cards, UPI, bank redirect)
- POST `/api/orders/<id>/stripe/create/` — creates intent, returns client_secret + publishable_key
- POST `/api/orders/<id>/stripe/confirm/` — server-side confirmation after 3DS/redirect
- Webhook: POST `/api/webhooks/stripe/` — HMAC signature-verified via Stripe-Signature header
- Handles payment_intent.succeeded and charge.refunded / payment_intent.refunded events
- @stripe/react-stripe-js Elements embedded inline on the order page

PayPal

- PayPal v2 REST Orders API — create !' capture flow

- Webhook: POST /api/webhooks/paypal/ — verify-webhook-signature via PayPal SDK
- Handles PAYMENT.CAPTURE.COMPLETED and PAYMENT.CAPTURE.REFUNDED
- 503 returned safely if PAYPAL_WEBHOOK_ID not configured

Cash on Delivery (COD)

- Admin-only mark-paid action: POST /api/orders/<id>/cod-mark-paid/
- COD orders excluded from online refund flow

8. Coupons & Discounts

- Percentage and fixed-amount coupon types
- Max discount cap for percentage coupons
- Minimum order amount requirement
- Global usage limit (total redemptions across all users)
- Per-user redemption limit
- Valid-from / valid-until date range
- Active / inactive toggle
- POST /api/coupons/apply/ — validates coupon + returns preview for current user
- Coupon code stored on order for audit trail
- Admin CRUD: create, update, delete via /api/coupons/

9. Product Reviews

- Authenticated users can submit one review per purchased product
- Star rating (1–5) + text body
- Admin moderation: approve / reject
- Aggregate rating and review count surfaced on product cards and detail page
- Paginated review list on product detail page

10. Returns & Refunds

Return Requests (Customer)

- POST /api/returns/ — submit return for delivered order
- Select items and quantities to return
- Return reasons: defective, wrong item, not as described, no longer needed, other
- RMA number auto-generated on submission
- Return request UI at /dashboard/orders/<id>/return

Refunds (Admin)

- POST /api/orders/<id>/refund/ — partial or full refund
- Stripe: Stripe.create_refund() with optional amount
- PayPal: refund-capture API call
- COD orders excluded (manual refund indicated via notes)
- Refund amount, date, and reference stored on order
- Refund email dispatched to customer via Celery task

11. Shipping

- Zone-based shipping cost computation (state + country + order subtotal)
- Shipping cost and estimated delivery range included in checkout quote
- Free-shipping threshold configurable via shipping zone rules
- Admin tracking update: POST /api/orders/<id>/tracking/
- Sets tracking_number, courier_name, tracking_url, estimated_delivery_date
- Order status transitions to shipped automatically on tracking save
- shipped_at timestamp recorded; delivered_at set when status moves to delivered

12. Email Notifications (Async)

- Order confirmation email dispatched via Celery on order creation
- Order status update email dispatched on every status transition
- Refund notification email with amount and reference
- Low-stock alert email to admins when product stock crosses threshold
- Forgot-password / reset-password emails
- All tasks use bind=True, max_retries=3, default_retry_delay=30s
- CELERY_TASK_ALWAYS_EAGER=True in DEBUG mode (no broker needed in development)

13. Django Backend Admin

- Obscure admin URL via DJANGO_ADMIN_URL environment variable
- All models registered with list_display, search_fields, filters
- Order admin: inline order items, status update, coupon info
- Product admin: image inlines, stock management, CSV import
- Coupon admin: usage stats, date range display
- Review admin: approve/reject actions
- Return admin: status management, admin notes
- Contact messages admin: status tracking

14. Storefront Admin Shell (/admin)

- Protected by AdminGuard — requires is_staff=True, redirects others
- Responsive sidebar layout with route-aware active state

Dashboard

- Revenue (paid orders), total orders, pending orders
- Total customers, new customers last 30 days
- Total products, active products, featured products, total categories
- Low-stock count, out-of-stock count
- New support messages count, newsletter subscribers
- Last-30-day orders and revenue cards

Orders

- Paginated orders table with status filter chips
- Per-order detail page with full item breakdown and totals
- Status dropdown — instantly updates order status via API
- Tracking & shipment form (tracking #, courier, URL, ETA) — marks as shipped
- Refund form — partial or full, dispatched to correct gateway

Products

- Product list with SKU, name, category, price, stock status

- One-click CSV export (date-stamped download)
- One-click CSV import (file picker !' multipart upload !' toast result)

Customers

- Paginated customer list with live search (email / name)
- Joined date, phone, full name columns

Coupons

- Create coupon: code, type, value, min order, per-user limit, validity dates
- Coupon list: discount, usage, validity, active badge
- Delete coupon with confirmation

15. Product Comparison

- Zustand store with localStorage persistence (solar.compare.v1)
- Maximum 4 products compared at once
- Toggle add/remove from any product card
- Compare badge in header showing count
- /compare page with side-by-side attribute table
- Attributes compared: price, brand, capacity, warranty, installation, stock
- Remove individual products or clear all

16. Search

- Search bar in ShopHeader (desktop + mobile drawer)
- Submit redirects to /products?search=<query>
- /search route acts as a redirect helper for deep-linked search URLs
- Backend full-text search across product name, brand, category

17. Contact & Newsletter

- Contact form: name, email, phone, subject, message
- Server-side validation and spam protection via throttle
- Contact messages stored in DB with status tracking (new / read / replied)
- Newsletter subscription: email capture with unsubscribe support
- Newsletter subscriber count exposed in admin dashboard

18. UI & UX Components

Core Components

- Button — size/variant system (primary, outline, ghost), loading spinner state
- FormField — label + input/textarea with error display
- Skeleton — product card skeleton, grid skeleton, table row skeleton
- EmptyState — dashed-border card with icon, title, description, action CTA
- Breadcrumbs — with embedded JSON-LD BreadcrumbList structured data
- Timeline — order status timeline with done / current / upcoming states
- PriceRangeSlider — dual thumb native range with filled track, accessible
- CookieBanner — GDPR-style consent banner persisted in localStorage

Shop Components

- ProductCard — image, badge, rating, wishlist toggle, add-to-cart
- ProductGallery — primary image + thumbnail strip
- AddToCartButton — stock-aware, quantity selector
- WishlistButton — toggle with optimistic update
- ReviewsSection — paginated reviews with star display
- ShopHeader — sticky, search bar, cart/wishlist badges, compare badge, admin link
- PayPalCheckout — SDK-loaded, capture flow with error handling
- StripeCheckout — Elements-embedded card/UPI with confirmPayment flow

19. SEO & Discoverability

- Root layout metadata: title template, description, OpenGraph siteName, robots
- metadataBase set from NEXT_PUBLIC_SITE_URL for absolute OG image URLs
- /sitemap.xml — static pages + all product slugs, cached for 1 hour
- /robots.txt — disallows /dashboard/, /admin/, /checkout/, /api/
- BreadcrumbList JSON-LD on pages using the Breadcrumbs component
- Canonical URL via metadataBase
- Cookie consent banner for GDPR/privacy compliance notices

20. Security

- JWT authentication — short-lived access tokens, refresh via HttpOnly or bearer
- Stripe webhook HMAC signature verification (Stripe-Signature header)
- PayPal webhook verify-webhook-signature SDK call
- CORS configured via CORS_ALLOWED_ORIGINS — no wildcard in production
- Rate limiting: AuthRateThrottle on login/register, UserRateThrottle on coupon apply
- Admin URL obscured via DJANGO_ADMIN_URL environment variable
- is_staff / is_installer flags read-only — no self-elevation via API
- SQL injection prevention via ORM parameterised queries
- XSS prevention via React's JSX escaping + Content-Security-Policy headers (nginx)
- File upload validation on CSV import (content-type + extension check)
- Input validation via DRF serializers (backend) and Zod (frontend)
- No secrets in source code — all sensitive values via environment variables

21. Performance

- React Query (TanStack Query) — stale-while-revalidate caching for all API calls
- Zustand persistent stores (cart, wishlist, compare) for instant reads
- Next.js static generation for marketing pages; dynamic for personalised routes
- Streaming CSV export (generator-based) to avoid memory spikes on large catalogues
- Redis cache with locmem fallback in development
- Celery for I/O-heavy tasks (email) off the request path
- next/image automatic WebP conversion and lazy loading
- Skeleton loading states on every async page to avoid layout shift

22. DevOps & Deployment

- Docker containers for Django (gunicorn) and Next.js

- docker-compose.yml for local full-stack dev (Django + Next.js + Nginx)
- Nginx reverse proxy with SSL termination (nginx.ssl.conf)
- Redis service and Celery worker service definitions for production compose
- Environment-based configuration — no hardcoded secrets
- Django manage.py migrate + collectstatic wired into entrypoint.sh
- Postgres in production, SQLite in development/test
- AWS S3 media storage switchable via USE_S3=true
- Stripe + PayPal webhooks fully documented in PRODUCTION_DEPLOY.md
- Pre-launch checklist: HTTPS, HSTS, DJANGO_DEBUG=False, superuser, email DKIM

23. Test Coverage

- 131 automated backend tests (pytest-django) — all passing
- Test modules: users, products, orders, coupons, reviews, wishlists, contacts
- Test modules: payments (Stripe + PayPal mock), security, password reset
- Factory-based test fixtures via factory_boy
- SQLite used for test runs (zero infrastructure, fast)
- CELERY_TASK_ALWAYS_EAGER=True ensures email tasks run synchronously in tests
- Frontend TypeScript strict build — npx tsc --noEmit clean
- Next.js production build — npm run build clean, 34 routes

